Wheat

Botanical name : *Triticum aestivum*

Origin and Area of cultivation

Earliest evidence for wheat cultivation comes from Fertile Crescent region. The common cultivated wheat, *Triticum aestivum* is cultivated for about 7,500 years. Wheat is mostly cultivated in the North Indian states such as Uttar Pradesh, Punjab, Haryana, Rajasthan, Madhya Pradesh and Bihar.

Uses

Wheat is the staple food in Northern India. Wheat flour is suitable to make bread and other bakery products. Processed wheat flour, that has little fibre, is called Maida which is used extensively in making parota, naan and bakery products. Malted wheat is a major raw material for producing alcoholic beverages and nutritive drinks.

Maize / Corn

Botanical name : *Zea mays*

Origin and Area of Cultivation

Maize is the only cereal that has originated and domesticated from the New World. Madhya Pradesh, Himachal Pradesh and Punjab are the major maize producing states of India. Whereas Perambalur, Ariyalur, Cuddalore, Dindigul and Tirupur are the major maize growing belts in Tamil Nadu.

Why do popcorn pops?



Endosperm in corn consists of two type namely soft and hard. In popcorn soft endosperm constitutes most part of the grain surrounded by thin layer of hard endosperm. When heated, the internal starch and protein are converted into gelatinous substances and when pressure mount further, the soft endosperm expands and explodes reversing the grain and the gelatinous starch are converted into foam, which readily solidifies outside and convert into crispy, tasty popcorn.



PSEUDO-CEREAL



Pseudo cereal - Chenopodium quinoa

The term pseudo-cereal is used to describe foods that are prepared and eaten as a whole grain, but are botanical outliers from grasses. Example: **quinoa**. It is actually a seed from the *Chenopodium quinoa* plant belongs to the family Amaranthaceae. It is a gluten-free, whole-grain carbohydrate, as well as a whole protein (meaning it contains all nine essential amino acids) and have been eaten for 6,000 years in Andes hill region.

Uses

Most of the corn produced is used as fodder than food. Corn syrup is used in the manufacture of infant foods. Corn is a raw material in the industrial production of alcohol and alcoholic beverages.

10.1.2 Millets (Siru Thaniyangal)

The term millet is applied to a variety of very small seeds originally cultivated by ancient people in Africa and Asia. They are gluten free and have less glycemic index.

Pearl Millet

Botanical name: *Pennisetum americanum*

Origin and Area of Cultivation

It is one of the millets introduced in India and Africa. Pearl Millet is rich in fibre, iron and minerals, stable food grain in many parts of India, especially in Gujarat and Rajasthan.

Uses

It is commonly used to make flat bread, gluten free cereal based products, porridge (Kambang koozh), biscuits, pasta and non-dairy probiotic beverages.

TN_GOVT_BOTANY_XII_PAGES_216-279 CH 10.indd 218 28-12-2022 15:38:16